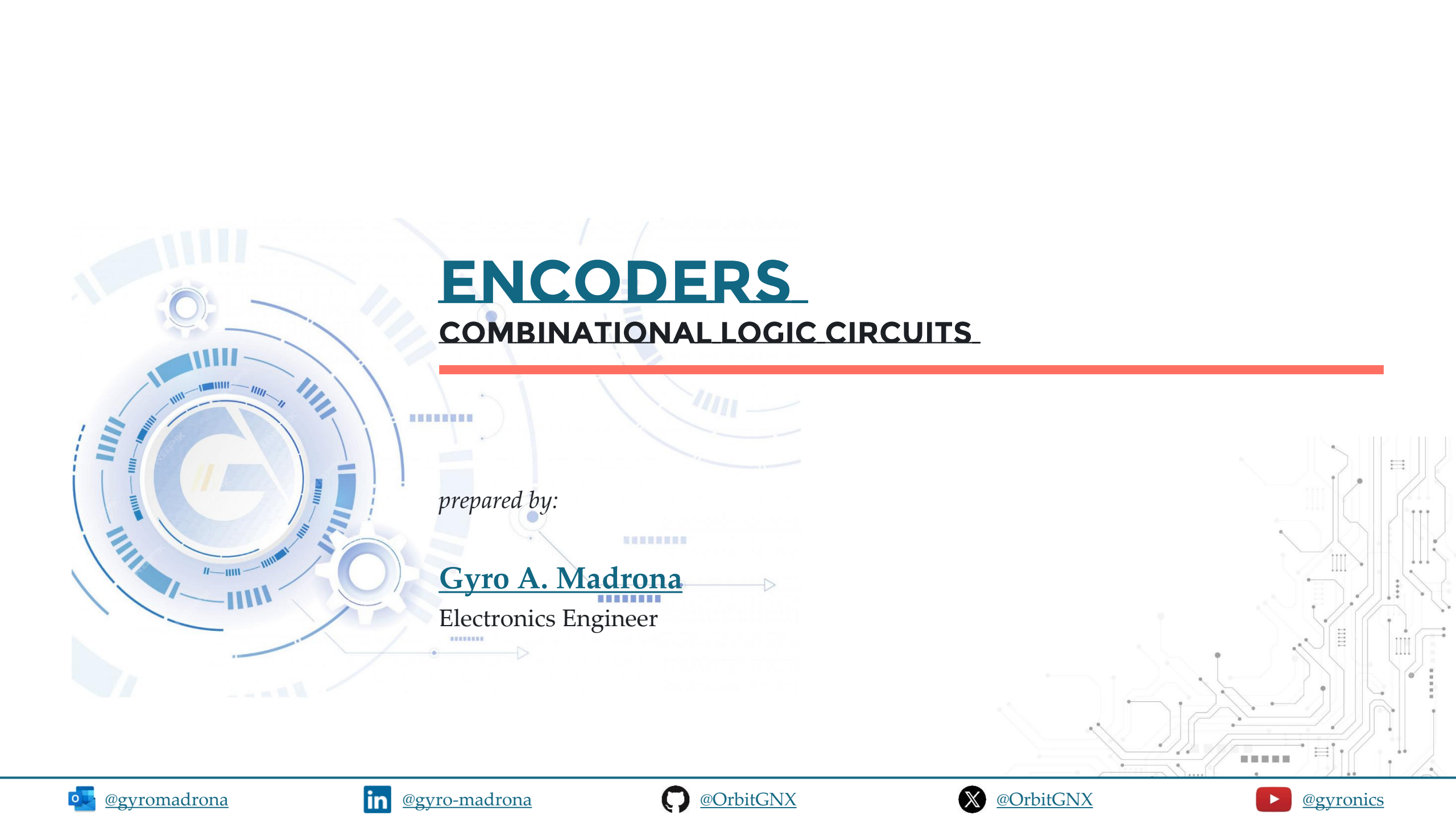




ENCODERS

COMBINATIONAL LOGIC CIRCUITS

prepared by:

Gyro A. Madrona
Electronics Engineer

TOPIC OUTLINE

Binary Encoder

Priority Encoder

Decimal-to-BCD Encoder

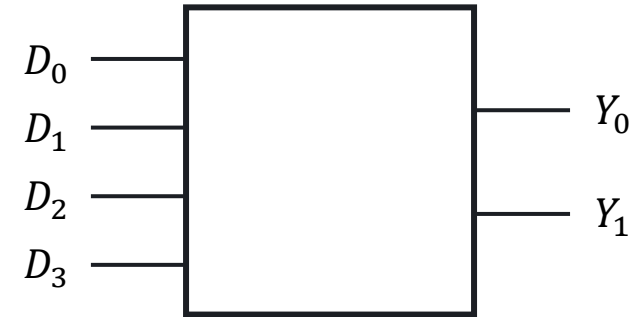


ENCODERS



BINARY ENCODER

Graphical Symbol



A binary encoder encodes information from 2^n inputs into an n -bit code.

Truth Table

D_3	D_2	D_1	D_0	Y_1	Y_0
0	0	0	1	0	0
0	0	1	0	0	1
0	1	0	0	1	0
1	0	0	0	1	1



EXERCISE

Implement a 4-to-2 binary encoder based on its truth table.

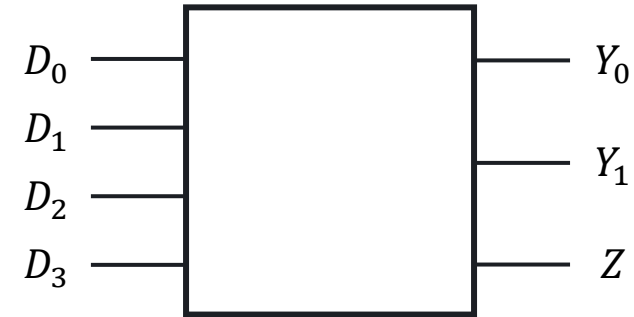
Solution

D_3	D_2	D_1	D_0	Y_1	Y_0
0	0	0	1	0	0
0	0	1	0	0	1
0	1	0	0	1	0
1	0	0	0	1	1



PRIORITY ENCODER

Graphical Symbol



A priority encoder assigns a binary code to the highest-priority input signal and ignores other inputs with lower priority.

Truth Table

D_3	D_2	D_1	D_0	Y_1	Y_0	Z
0	0	0	0	d	d	0
0	0	0	1	0	0	1
0	0	1	x	0	1	1
0	1	x	x	1	0	1
1	x	x	x	1	1	1



EXERCISE

Implement a 4-to-2 priority encoder based on its truth table.

Solution

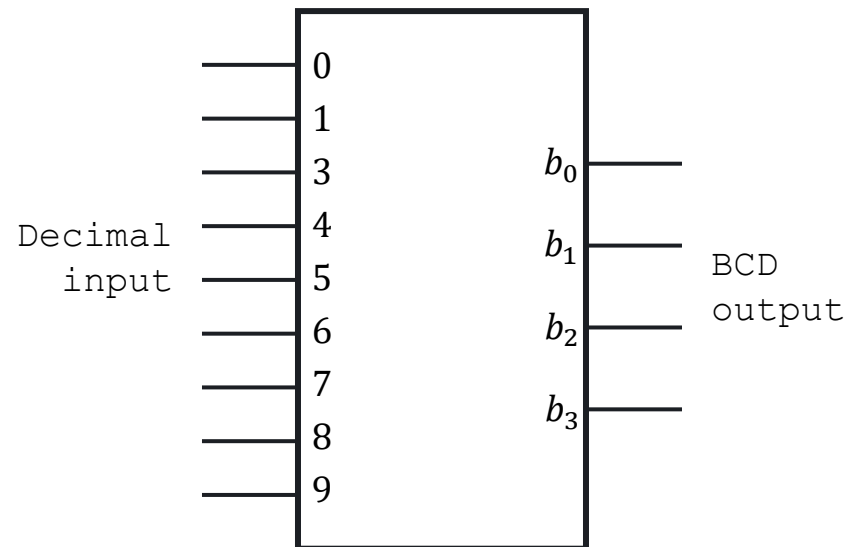
D_3	D_2	D_1	D_0	Y_1	Y_0	Z
0	0	0	0	d	d	0
0	0	0	1	0	0	1
0	0	1	x	0	1	1
0	1	x	x	1	0	1
1	x	x	x	1	1	1



DECIMAL-TO-BCD ENCODER

The decimal-to-BCD encoder has ten inputs – one for each decimal digit – and four outputs corresponding to the BCD code.

Graphical Symbol



Truth Table

Decimal	BCD Code			
	b_3	b_2	b_1	b_0
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1

EXERCISE

Implement a decimal-to-BCD encoder based on its truth table.

Solution

Decimal	BCD Code			
	b_3	b_2	b_1	b_0
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1



LABORATORY

